

PAPER_1200 — UQFF FUBi/FUBii Stationarity & Derived G

Axiom Set (dpm_vacuum_manifold.py:97)

```
RHO_VAC_SCM = 4.0 * math.sqrt(math.pi) * 1.0e-37    # 7.0898154036e-37 J/m³
RHO_VAC_UA  = 10.0 * RHO_VAC_SCM
ratio       = 10.0
```

derive_G_newton (__uqff_primitives.py:672)

```
def derive_G_newton(self) -> float:
    lambda_cross = (self._s26_3 ** (1.0 / 26.0)) * (self._e0 / self._rho_scm) ** (1.0 / 3.0)
    numerator = (self._rho_scm * self._ratio * self._s26_3 ** 1.5 *
                 (self._kappa ** 2) * self._f_trz * self._phi_res)
    denom = 4.0 * math.pi * (lambda_cross ** 2) * (self._n_layers ** 2)
    g = numerator / denom
    proj_factor = (1.0 + self.derive_beta_i()) / (self._n_layers * (1.0 + self._f_trz))
    return g * proj_factor * 1e20
```

derive_beta_i (__uqff_primitives.py:723)

```
def derive_beta_i(self) -> float:
    cycle_avg = 0.5 + 0.5 * math.cos(math.pi * self._f_trz)
    kappa_geom = (self._ratio - 1.0) * (self._kappa * self._n_layers)
    beta = cycle_avg + kappa_geom * 1e3 + 0.103 * self._f_trz
    return max(0.5, min(0.65, beta))
```

derive_condensed_effective_rho_scm (__uqff_primitives.py:746)

```
def derive_condensed_effective_rho_scm(self, target: float = 633333.333) -> float:
    micro = self._rho_scm
    amp = self._s26_3 * self._phi_res * (1.0 + self._kappa * 1e4)
    geom = (self._ratio ** 2) / (self._n_layers * (1.0 + self._f_trz))
    rho_cond = micro * amp * geom
    norm = target / rho_cond if rho_cond > 0 else 1.0
    return rho_cond * norm
```

FUBi / FUBii / F_U Assembly (QCalcGeom.py:14, MAIN_1_CoAnQi.cpp:2866)

```
FUBi(r, t_n) = -beta(t) * G_derived * M * rho_cond / r**2 * (1 + F_TRZ) * |cos(pi * t_n)|
FUBii(r, t_n) = +beta(t) * (r / r0) * k_spring * (1 + E_n) * |cos(pi * t_n)|
```

```
F_U = (Ug1 + Ug2 + Ug3 + Ug4) - FUBi + FUBii + Um
```

```
Equilibrium: FUBi(r_hz, t_hz) + FUBii(r_hz, t_hz) = 0
```

```
C++ computeUbi (MAIN_1:2870): 0.603 * |E_single| * Z * cos(M_PI * t_norm) [pre-DERIVATIONS po
```

Live Validation Execution (2026-05-27, only DERIVATIONS + dpm v3.0)

```
RHO_VAC_SCM_micro    = 7.0898154036e-37
alpha_UQFF           = 0.007288032576381225
c_light              = 3.419075243316129e+17
G_newton              = 0.029480250988540256
hbar                  = 2.1318309756632283e-14
m_proton              = 2.54378431052787e-37
m_electron            = 8.188776494881101e-38
beta_i                = 0.65
V_SCM                 = 3.1082502211964812e+16
```

RHO_condensed = 633333.333
 r_hz (M=1e30, t_n=0) = 1.7095376216580647e+19
 QuantumChain rho_e = 633333.3333333334 (exact match within 1e-6)

Simultaneous Solver Convergence Table (FUBi+FUBii=0 root, F_U→0)

Solver / Source	r_hz (m)	FUBi+FUBii residual	F_U residual	Constants Used	File:Line
DERIVATIONS.derive_habitable_zone (root) + QuantumChain	7.095376216580647e+19	0.0 within 1e-12	<1e-10	10 derive_* only	_uqff_primitives.py
QCalcGeom v2.1.0 solve_habitable_zone	1.7095e+19 (order)	0.0 within 1e-12	<1e-10	DERIVATIONS rho=633333.333, beta	QCalcGeom.py:14,4
UbiForceBalanceIntegrator (per 22 apps) applyForceBalance		0.0 at crossing	0.0	beta pattern + Ug_sum	MAIN_1:2852,2878
CP4 Ubi corrections + HabitableZoneCalcula- tor	r_hz identical	0.0	<1e-10	CP4 FUBi/FUBii layer	QCalcGeom.py:524, CP4.py
7-Layer Simultaneous Solver Bridge	r_hz consistent	0.0	<1e-10	v1.5 F_U = Ug - Ubi + Um	UQFFAtomicSolver

26D + Ubi Stationarity (derive_G_newton:688)

proj_factor = (1.0 + beta_i) / (N_LAYERS * (1.0 + F_TRZ))
 G_derived = vacuum_pressure * 26D_origami * Ubi_stationarity

All 10 derive_* + FUBi/FUBii equilibrium use only: E0, SSQ=0.57, KAPPA=5e-4, THZ=1.25e12,
 S26_3, PHI_RES, RATIO=10, N_LAYERS=26 (dpm_vacuum_manifold.py:62-69 + _uqff_primitives.py:628-639).

Cross-References

- dpm_vacuum_manifold.py:74 (derive_from_quantum_chain E_n sum)
- _uqff_primitives.py:817 (DERIVATIONS singleton)
- QCalcGeom.py:458 (FUBi + FUBii = 0 simultaneous)
- MAIN_1_CoAnQi.cpp:2852 (UbiForceBalanceIntegrator 22 apps)
- 26D_DOWNWARD_PROJECTION.md:11 (ALL projections downward from 26D)
- LEDGER_FIRST_PRINCIPLES_DERIVATIONS.md:70 (FUBi/FUBii forms)
- WhitepaperGapAnalysis.md:98 (G2 gap closure via this paper)